

1 Foundational Concepts of Analytics

Business Analytics

Is the practice of using data, statistical methods, and technology to analyze past performance and guide future decisions.

Example: A retail chain might analyze sales to determine which products to stock more of each season.

2: Data Analytics

Is a process of examining raw data to uncover patterns, draw conclusions and support decision making.

Example: A streaming service might analyze viewer watch history to recommend personalized shows.

3 Business Intelligence (BI)

Is the technology-driven process of collecting, storing, and analyzing data to provide historical, current, and predictive views of operations.

Example: A logistics company uses BI dashboard to track delivery times.

4 Data-driven decision making (DDDM)

Is a management philosophy that emphasizes on using actual data, metrics and facts, to drive strategic business choices rather than relying on intuition or experience.

Example: an e-commerce company might test two website designs with live traffic and choose a version that generates more sales based on click-through data.

5. Big Data

Refers to extremely large complex datasets that cannot

easily be managed or processed with traditional data tools, often characterized by high volume, velocity and variety.
 Example: A social media platform like Facebook analyzes billions of daily posts, likes, and shares to personalize news feeds and target advertisements.

6. Descriptive Analytics

Is a process of summarizing historical data to understand what has happened in a business or organization.

Example: A university might use descriptive analytics to calculate the total number of students each semester and their average graduation rates.

7. Diagnostic Analytics

Examines data to understand why something happened by identifying patterns, relationships, or root causes behind past outcomes.

Example: A company noticing a drop in sales might analyze customer demographics and website traffic to discover that a recent price increase drove away younger buyers.

8. Predictive Analytics

Uses historical data, statistical models, and machine learning techniques to forecast future outcomes or identify the likelihood of specific events occurring.

Example: A credit card company might ~~also~~ analyze past spending and payment behaviour to predict which customers are most likely to miss a future payment.

9 Prescriptive Analytics

Prescriptive Analytics goes beyond predicting future outcomes by recommending specific actions or decisions to take, often using optimization, simulation or decision support algorithms.

Example: A delivery company can use prescriptive analytics to determine the most fuel-efficient routes for its trucks in real-time, balancing delivery windows and traffic conditions.

10 Data Literacy

Is the ability to read, understand, create and communicate with data as information, enabling individuals to ask appropriate questions and interpret analytical results correctly.

Example: a ~~marketing~~ marketing manager with data literacy can review a sales report, identify misleading averages and explain why the team should focus on median customer spending instead.

Data, Measurement & Reporting

11 Structured Data

Refers to information that is organized predefined, format, typically row and columns, making it easy to search and analyze by databases or spreadsheets.

Example: An ~~excel~~ spreadsheet containing customer names, phone numbers, and purchase dates is structured data.

12 Unstructured Data

Is information that does not follow a predefined format or organization structure, making more difficult to

collect, process, and analyze using traditional databases. Example: customer reviews, social media posts, video files, or email messages are all forms of unstructured data.

13 Data Warehouse

Is a centralized system that stores large amounts of historical data from multiple sources, organized for analysis and reporting rather than daily transactions.

Example: A hospital chain might use a data warehouse to combine patient records, billing information, and lab results from all its locations to identify trends in readmission rates.

14 Data Mining

Is a process of discovering hidden patterns, correlations, and useful insights within large datasets by using statistical techniques and machine learning algorithms.

Example: A grocery store might use data mining to analyze transaction records and find that customers who buy diapers are also likely to purchase beer on Friday evenings.

15 Data Visualization

Is a graphical representation of information and data using visual elements like charts, graphs, maps, and dashboards to help people understand patterns, trends, and outliers.

Example: A sales manager might use a line chart to show monthly revenue over a year, instantly revealing seasonal peaks and dips.

16 Dashboard

Is a visual display that consolidates and presents key performance indicators (KPIs) and important metrics in a single, easy-to-read interface for real-time monitoring.
 Example: An e-commerce company might use a dashboard to show daily website visitors, conversion rates, and total sales revenue all on one screen.

17 Metric

A Metric is a quantifiable measure used to track, assess, and communicate performance or progress toward a specific business goal.

Example: The percentage of customers who stop using a service within a given period, is a common metric for subscription-based business.

18 Key Performance Indicator (KPI)

Is a measurable value that demonstrates how effectively an organization is achieving key business objectives, linking performance directly to strategic goals.

Example: In Online retail store, the "cart abandonment rate" is a key KPI that shows the percentage of shoppers who add items to their cart but leave without completing the purchase.

19 Benchmark

Is a standard or point of reference against which performance processes, or results can be compared and measured.

Example: a small restaurant might compare its average customer wait time to the industry benchmark of 10

minutes to see if it needs to improve kitchen efficiency

20. Segmentation

Is a process of dividing a broad group of customers, dates, or markets into smaller subsets based on shared characteristics like demographics, behaviour or needs.

Example: A clothing retailer might segment its customers into group such as "budget conscious", "professional worker", "luxury brand" to tailor marketing campaigns

Customer & Sales KPIs

21 Conversion Rate

Is the percentage of users who complete a desired action out of the total number of visitors or opportunities, measuring how effectively a process turns interest into results.

Example: If an e-commerce website receives 1000 visitors in a day and 50 of them make a purchase, the conversion rate is 5%.

22 Customer Acquisition Cost (CAC)

Is the total cost a business incurs to acquire a new customer, including sales expenses and marketing over a specific period.

Example: If a company spends ₹5000 on marketing in one month and gains 100 new customers, its CAC is ₹50 per customer

23 Customer Retention Ratio Lifetime Value (CLV/LTV)

Is a prediction of the total profit a business can expect from a single customer the entire duration of their relationship.

Example: If a coffee shop customer spends R5 per visit, comes twice a week for five years. The estimated CLV is roughly R 2600.

24 Churn Rate

Is the percentage of customers who stop using a company's product or service over a period, indicating customer retention and satisfaction.

Example: if a streaming service starts a month with 1000 subscribers and ends with 950, the churn rate is 5% for that month.

25 Customer Retention Rate

Is the percentage of customers a business keeps over a given period, measuring loyalty and the effectiveness of retention strategies.

Example: If a fitness club starts a year with 500 members, loses 50, but adds 100 new ones, its retention rate would be 90% (450 of the original remained).

26 Net Promoter Score (NPS)

Is a simple metric that measures customer loyalty by asking one question: "On a scale of 0-10, how likely are you to recommend our company to a friend?"

Example: A Rating system score and customer satisfaction.

27 Average Order Value (AOV)

Average Order Value is the average amount a customer spends per transaction on a website or in a store, calculated by dividing total revenue by number of orders.

Example: if an online bookstore earns R10 000 from 200 orders in a week, the AOV is R50 per order

28 Sales Funnel

Is a visual model that represents the customer's journey from first becoming aware of products to completing a purchase, with each stage narrowing down as people drop off.

Example: an online course provider might have 10000 website visitors (awareness), 1000 who sign up for free trial (interest), and 100 who buy the full course (purchase) forming a funnel.

29 Marketing Campaign

Is a coordinated series of promotional activities and messages designed to achieve a specific goal.

Example: A coffee shop might run a summer campaign offering "Buy 5 drinks, get 1 free" through social media and store posters to boost repeat visits.

30 Post-Campaign Analysis

Is the process of evaluating the performance and results of a completed marketing campaign key metrics against the original goals.

Example: after a holiday email promotion, a retailer might analyze open rates, click-through rates, and total sales generated to determine if the campaign met its revenue target.

Incremental Revenue

Is the additional income generated from a specific business activity, such as marketing campaign, new product launch, or pricing change compared to what would have been earned

without it.

Example: if a retail store normally makes R10000 per week but runs a promotion and makes R13000, the incremental revenue from promotion is R3000.

3.2 Return on Investment (ROI)

Is a performance measure used to evaluate the efficiency or profitability of an investment, calculated by dividing net profit by initial cost.

Example: if a company spends R5000 on an online ad campaign and earns R15000 in sales directly from it, the ROI is 200%.

3.3 Return on Ad spend (ROAS)

Is a marketing metric that measures the total revenue generated for every dollar spent on advertising, calculated by dividing revenue from ads by the cost of ad cost.

Example: if a company spends R1000 on a social media campaign and earns R5000 in sales, the ROAS is 5:1 meaning R5 returned for every R1 spent.

3.4 Click-Through rates (CTR)

Is a percentage of people who click on a specific link ad, or email call-to-action out of the total number who view it.

Example: if an online banner ad receives 1000 impressions and 50 people click on it, then the CTR is 5%.

3.5 Bounce Rate

Is the percentage of website visitors who land on a

page and then leave without clicking anything, navigating to another page, or taking any action

Example: if 200 people visit a blog post and 80 close the tab immediately without reading or clicking a link, the bounce rate is 40%

36 A/B Testing

Is a method of comparing two versions of a webpage, email, or ad to see which one performs better by showing each to a random half of visitors

Example: an online store might test a buy now red or green button, if one gets more clicks, then it is the winning version

37 Lead Generation

Is a process of attracting and capturing interest from a potential customer (lead) who may ~~make~~ convert and make a purchase, typically by offering something valuable in exchange for their contact information.

Example: A software company might offer a free book or webinar, and when visitors provide their email address to download it, they become a lead.

38 Marketing Attribution

Is the practise of assigning credit or value to the different marketing channels that influence a customer's decision to convert.

Example: If a customer sees a FB ad, later clicks on the email link, and finally makes a purchase through a google search, attribution modeling helps determine how much credit each touchpoint receives

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